Trajectory Design and Link Selection in UAV-Assisted Hybrid Satellite-Terrestrial Network

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***Abstract*— Mobile relaying is envisioned as a promising technology to alleviate the masking effect in satellite links. Nevertheless, due to the mobility and limited communication range, the network topology in mobile relay networks becomes highly dynamic and time-varying, resulting in increased difficulty of network optimization. This letter investigates unmanned aerial vehicles (UAVs) aided hybrid satellite-terrestrial network, where UAVs are served as mobile relay base stations to assist the communication between the satellite and ground users. We aim to maximize the number of served users by optimizing UAV trajectory and user link selection. To solve the formulated mixed-integer and non-convex optimization problem, we first find the optimal link selection via a designed graph neural network (GNN), and then adjust the UAV locations by using model-free reinforcement learning (RL), alternately. Numerical results demonstrate that our proposed scheme is superior to the state-of-the-art RL algorithms.**

***Index Terms*— Unmanned aerial vehicles (UAVs), trajectory design, graph neural network (GNN), reinforcement learning, satellite networks.**

I. Introduction

THE sixth-generation (6G) system aims to provide seamless connectivity and high-speed broadband access through the integration of satellite communications (SatCom) and the existing terrestrial cellular networks. However, the satellite coverage is limited by the masking effect resulting from atmospheric conditions (e.g., rain) and terrestrial barriers (e.g., mountains). This issue is particularly serious in the case of high-frequency bands like Ka-band, which is widely considered in the satellite-terrestrial communication.

With the three-dimensional (3D) mobility features, unmanned aerial vehicles (UAVs) can be served as mobile relay base stations, which has been recognized as a promising solution to overcome the deleterious masking effect in hybrid satellite-terrestrial networks (HSTN). In recent years, UAV-assisted HSTN have attracted much attention from researchers [1], [2]. In [1], the outage probability of an HSTN with 3D mobile UAV relaying is analyzed under the random waypoint model. The work in [2] has investigated the secrecy performance of different UAV relay selection strategies in the presence of an aerial eavesdropper. However, both of them consider that the UAV relays are deployed in a fixed cylindrical region with stochastic UAV mobility models. To the authors' best knowledge, the trajectory design in the UAV-assisted HSTN has not been fully investigated and needs the utmost attention prior to deploying UAVs in HSTN.

The design of UAV relay network in an HSTN scenario is more challenging compared with traditional networks. First, the UAV trajectory design should take the mobility of satellites into consideration. This is particularly important for the low-earth orbit (LEO) based satellite networks, in which the service time during the satellite coverage is around 15 minutes. Second, ground users can be served by both terrestrial and UAV connections in HSTN, which is different from most of the existing studies on UAV trajectory design where only UAV connection is considered. Since the UAVs have limited energy resources and communication bandwidth, not every transmission request can be satisfied. To balance traffic loads in HSTN, it is essential and challenging to perform dynamic link selection in the face of unknown user mobility.

Motivated by the above research gaps, we focus on the system design for UAV-assisted HSTN and formulate a joint trajectory design and link selection problem for maximizing the number of served users subject to the minimum downlink rate constraint. To address the non-convex nature of the formulated problem, we propose a two-stage optimization scheme. Specifically, we first find the optimal link selection for each transmission request via graph neural networks (GNN). Since the interactions between non-terrestrial and terrestrial nodes are quite different, it is beneficial to adopt GNN models that capture how these interactions influence the outcome. This is different from most existing works on learning-based UAV relay networks which directly learn the actions that lead to high reward. Then, we learn the optimal UAV trajectory by using model-free reinforcement learning (RL). Numerical results show that the proposed scheme can effectively improve the performance of UAV-assisted HSTN compared with the state-of-the-art RL algorithms.

II. System Model and Problem Formulation

Consider a downlink transmission system, where S satellites communicate with N ground users via K decode-and-forward (DF) UAV relays, as shown in Fig. 1. The UAVs use the satellite link as backhaul to the core internet. Meanwhile, ground users can be served by a terrestrial macro base station (MBS) located at the center of the geographical area, whose coordinate is (x_b, y_b). For the downlink transmission, each user can obtain the requested contents via the following two ways.

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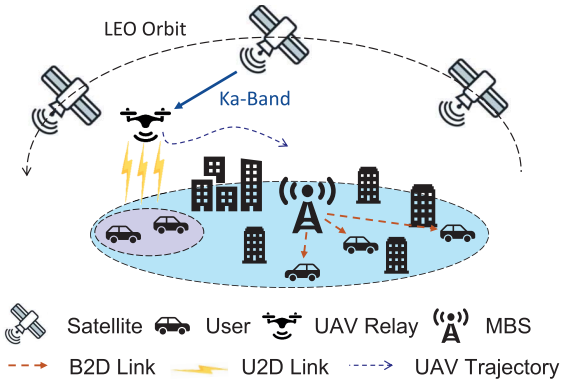


Fig. 1. Illustration of the considered UAV-assisted HSTN.

- **MBS-to-device (B2D) link:** The MBS directly transmits data from the core network to the ground users over the sub-6 GHz band.
- **UAV-to-device (U2D) link:** The LEO satellite first transmits data from the core network to the UAVs over Ka-band. Then, the UAVs forward the received data to the ground users over the sub-6 GHz band.

Additionally, the total bandwidth of the cellular network is divided into different subchannels shared among users to avoid mutual interference.

We evenly discretize the time horizon into T time slots. At each time slot, the UAVs can hover or move vertically or horizontally with a maximum speed V_{UAV}. Let $h_{k,t}^{b} \in [h_{\min}, h_{\max}]$ denote the altitude of the k_{th} UAV at the t_{th} time slot, where $h_{\min}$ and $h_{\max}$ are the minimum and the maximum altitudes that the UAV can reach, respectively. We consider that the MBS serves as a UAV charging station for the safe and continuous operation of the UAV network, as considered in [3]. Thus, the UAVs' initial and final locations are the same as that of MBS. Moreover, at the t_{th} time slot, the locations of the s_{th} LEO satellite, the k_{th} UAV, and the n_{th} user are respectively denoted by $w_{s,t}^{\text{LEO}} = (x_{s,t}^{\text{LEO}}, y_{s,t}^{\text{LEO}}, h_{s,t}^{\text{LEO}})$, $w_{k,t}^{\text{UAV}} = (x_{k,t}^{\text{UAV}}, y_{k,t}^{\text{UAV}}, h_{k,t}^{\text{UAV}})$, and $w_{n,t}^{\text{U}} = (x_{n,t}^{\text{U}}, y_{n,t}^{\text{U}})$. To model the user movement, we adopt the Gauss Markov mobility model, in which the velocity and the moving direction of the n_{th} user at the present time slot are related to those at the previous time slot. Please refer to [4] for more details.

A. Channel Model

1) Satellite to UAV Link: We assume that the satellites have a circular orbit around the Earth. As in [5], the shadowed-Rician (SR) fading model is adopted to characterize the channel condition of land mobile satellite links. The analytical expression presented in [6] is used to characterize the Doppler shift caused by the movement of satellites, which is omitted here due to space limitation. Let $\phi_{s,k} \in [0, 2\pi)$ and $\zeta_{s,k}$ denote the stationary random phase and the deterministic phase of the LoS component, respectively. The corresponding channel fading coefficient between the s_{th} satellite and the k_{th} UAV is given by $\chi_{s,k} = A\exp(j\phi_{s,k}) + Z\exp(j\zeta_{s,k})$, where A and Z denote the amplitude of scattering and line-of-sight (LoS) component following the Rayleigh and Nakagami-m distributions, respectively. The SR fading distribution can be expressed as $\chi_{s,k} \sim \text{SR}(\Omega_{s,k}, b_{s,k}, m_{s,k})$, with $\Omega_{s,k}$ is the average power of LoS component, $2b_{s,k}$ is the average power of the scatter component, and $m_{s,k}$ is the Nakagami-m fading parameter. Then, the received signal-to-noise ratio (SNR) of the satellite-UAV link at the t_{th} time slot is $\Gamma_{s,k,t}^{\text{LEO}} = P_{\text{LEO}}\chi_{s,k}{d_{s,k,t}^{\text{LEO}}}^{-\gamma_1}/B_sN_0$, where P_{LEO} denotes the transmit power of the LEO satellite; N_0 is the noise power; B_s is the frequency resource allocated to the UAV. Moreover, ${d_{s,k,t}^{\text{LEO}}}^{-\gamma_1}$ is the large-scale path loss, where $d_{s,k,t}^{\text{LEO}}$ denotes the distance between the s_{th} satellite and the k_{th} UAV; γ_1 is the path loss exponent of the satellite-UAV link. Since the horizontal distance between the UAV and the satellite is relatively small compared to the height of the satellite, we can approximate $d_{s,k,t}^{\text{LEO}} \approx h_{s,t}^{\text{LEO}}$. Hence, the achievable data rate of the k_{th} UAV from the s_{th} LEO satellite, denoted by $C_{s,k,t}^{\text{LEO}}$, can be written as $C_{s,k,t}^{\text{LEO}} = B_s\log_2(1 + \Gamma_{s,k,t}^{\text{LEO}})$. As in [2], we consider that each UAV always chooses the satellite using the maximum SNR criterion, i.e., $s^*(t) = \arg\max_{s} \Gamma_{s,k,t}^{\text{LEO}}$. For simplicity, we assume that a seamless satellite handover protocol exists, which allows uninterrupted user's session once a user is connected to a UAV.

2) UAV to User Link: For the U2D link, we adopt a widely used log-normal shadowing channel [7] which takes the probabilistic line-of-sight (LoS) and non-line-of-sight (NLoS) links into consideration. We ignore the small-scale fading due to the lower probability of multipath fading in the air-to-ground channel. Let $\bar{l}_{k,n,t}$ denote the large-scale channel power gain between the k_{th} UAV and the n_{th} user, including both the path loss and shadowing effect. The expression of $\bar{l}_{k,n,t}$ is omitted here and all necessary details can be found in [8]. The SNR of the ground user at the t_{th} time slot is $\Gamma_{k,n,t}^{\text{UAV}} = \alpha_{k,n,t}P_{\text{UAV}}\bar{l}_{k,n,t}/W_nN_0$, where P_{UAV} is the transmit power of the UAV relay; $\alpha_{k,n,t} \in \{0, 1\}$ is the decision binary variable to indicate whether the n_{th} user has been served by the UAV; W_n is the allocated bandwidth to the n_{th} user. Assume that the total bandwidth available to the UAV is equally divided among the associated users. Thus, the channel capacity between the k_{th} UAV and the n_{th} user will be $C_{k,n,t}^{\text{UAV}} = W_n\log_2(1 + \Gamma_{k,n,t}^{\text{UAV}})$.

3) MBS to User Link: For the B2D links, the standard distance-based power-law path loss model and Rayleigh fading environment are considered. Let $g_{n,t}^{\text{BS}}$ denote the channel gain between the MBS and the n_{th} user at the t_{th} time slot, which is assumed to be independent and identically distributed exponential random variables with mean μ_g. Therefore, the SNR of the link between the MBS and the n_{th} user is $\Gamma_{n,t}^{\text{BS}} = \beta_{n,t}P_{\text{BS}}g_{n,t}^{\text{BS}}{d_{n,t}^{\text{BS}}}^{-\gamma_2}/B_nN_0$, where P_{BS} denotes the transmit power of the MBS; $d_{n,t}^{\text{BS}}$ is the distance between the MBS and the n_{th} user; γ_2 denotes the path loss exponent of the B2D link; $\beta_{n,t}$ is the decision binary variable to indicate whether the n_{th} user has been served by the MBS; B_n is the allocated bandwidth.

B. UAV Energy Consumption Model

We consider the propulsion energy consumption while ignoring the relatively small energy cost spent on transmitting

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and receiving. For a fixed-wing UAV in a steady straight-and-level flight (SLF), the instantaneous propulsion energy consumption at the $t_{\rm th}$ time slot can be calculated as [9]

$$E(V_{k,t}) = \begin{cases} E_0, & \text{if } V_{k,t}=0, \\ c_1\|V_{k,t}\|^3 + \frac{c_2}{\|V_{k,t}\|}\left(1+\frac{\|a_{k,t}\|^2}{g^2}\right), & \text{otherwise}, \end{cases} \quad (1)$$

where E_0 is the UAV hovering power; $V_{k,t}$ is the flying speed; c_1 and c_2 are modeling coefficients to account for the parasitic drag and lift-induced drag, respectively; g is the gravitational acceleration (9.8 m/s^2); $a_{k,t}$ is the acceleration of the UAV. Then, the accumulated energy consumption over t is given by $E^{\mathsf{fly}}_{k,t} = \sum_{t\in T} E(V_{k,t})\delta$, where δ is the slot duration.

C. Problem Formulation

The coverage region of the considered HSTN refers to the area where the SNR of the associated user is above a predefined threshold. Let $\eta_{\rm th}$ be the minimum SNR requirement for the ground users. The coverage indicator for user n depends on the UAVs' locations as well as the link selection decision, which is defined as:

$$U_n\left[\mathbf{w}^{\mathsf{UAV}}_{\mathbf{t}}, \mathbf{u_t}\right] = \begin{cases} 1, & \text{if } \Gamma^{\mathsf{LEO}}_{s^*,k,t} \geq \eta^{\mathsf{LEO}}_{\rm th}, \Gamma^{\mathsf{UAV}}_{k,n,t} \geq \eta_{\rm th}, \\ & \text{or } \Gamma^{\mathsf{BS}}_{n,t} \geq \eta_{\rm th}, \\ 0, & \text{otherwise}, \end{cases} \quad (2)$$

where $\mathbf{u_t} = \{\alpha_{1,1,t}, \ldots, \alpha_{K,N,t}, \beta_{1,t}, \ldots, \beta_{N,t}\}$ is the link selection vector, $\mathbf{w}^{\mathsf{UAV}}_{\mathbf{t}} = \{w^{\mathsf{UAV}}_{k,t}\}$, and $\eta^{\mathsf{LEO}}_{\rm th}$ is the SNR threshold for the satellite backhaul link. To support high number of users with the required SNR, the optimization of joint trajectory design and link selection is considered as

$$\max_{\mathbf{w}^{\mathsf{UAV}}_{\mathbf{t}}, \mathbf{u_t}} \quad \sum\nolimits_{t=1}^{T}\sum\nolimits_{n=1}^{N} U_n\left[\mathbf{w}^{\mathsf{UAV}}_{\mathbf{t}}, \mathbf{u_t}\right] \quad (3a)$$

$$\text{s.t.} \quad w^{\mathsf{UAV}}_{k,1} = w^{\mathsf{UAV}}_{k,T} = (x_b, y_b, 0), \forall k, \quad (3b)$$

$$E^{\mathsf{fly}}_{k,t} \leq E_{\mathsf{max}}, \forall k, \quad (3c)$$

$$h_{\mathsf{min}} \leq h^b_{k,t} \leq h_{\mathsf{max}}, \forall k, \quad (3d)$$

$$\sum\nolimits_{n=1}^{N} C^{\mathsf{UAV}}_{k,n,t} \leq C^{\mathsf{LEO}}_{s^*,k,t}, \forall k, \quad (3e)$$

$$\sum\nolimits_{k=1}^{K} \alpha_{k,n,t} + \beta_{n,t} \leq 1, \forall n, \quad (3f)$$

where E_{max} is the energy capacity for each UAV. Note that (3b) specifies the constraints on UAVs's initial and final locations; the constraint (3c) represents the energy constraint for each UAV; the constraint (3f) denotes the link scheduling constraint. Due to the presence of binary user association variables, the formulated problem is a mixed-integer programming problem with strong NP-hardness, which is difficult to be solved by the standard optimization techniques[1]. Therefore, we utilize RL based methods to learn the environment and solve the formulated problem.

[1]Although some heuristic methods may be able to solve this problem, they are known to suffer from slow convergence and therefore are not applicable for timely decision making in highly dynamic space-air-ground integrated network environments.

III. Proposed Algorithm

In the following, we propose a two-stage learning technique to solve this problem. The basic idea of the two-stage optimization is to fix the long-term link selection policy on the MBS side and optimize the short-term flight control policy on the UAV side. First, at the MBS side, the optimal link scheduling is predicted by a centralized graph attention network (GAT) model [10] with user distribution as input features. Second, a distributed UAV path planning is performed via the model-free Q-learning given the optimal link scheduling.

We model the HSTN as a time-varying graph and exploit the GAT for learning the cooperative link scheduling policies among users over the graph. Unlike traditional GNN methods which only consider static graphs, GAT works well in such a dynamic environment with moving nodes that can join/leave the system at any time. GAT can extract the hidden node-embedding feature by iteratively exchanging information with neighboring nodes. The node-embedding feature of each node is a function of the node's observation and action (link selection) as well as other nodes' contributions, which is employed to characterize the underlying network structure. Let $\mathrm{G}_t = \{\mathbb{V}_t, \mathbb{E}_t\}$ denote a connected undirected graph for $\mathbf{u_t}$, where $\mathbb{V}_t = \{v_{i,t}\}$ and $\mathbb{E}_t$ represent the vertex and edge sets, respectively. Denote (i,j) is an edge in $\mathbb{E}_t$ if node i can communicate directly with node j. Moreover, the set of neighbors of i is denoted by $N(i)$.

The computation step of our GNN framework in the training phase is illustrated in Fig. 2. First, we initialize the node and edge features by transforming the spatial distribution of HSTN nodes into a graph model. During the information exchange and aggregation phase, the calculation of the node-embedding features of node i at the $m_{\rm th}$ iteration, denoted as h_i, is given as follows [10]

$$h^{(m)}_i = \sigma\Big(\sum_{j\in N(i)} a^{(m)}_{ij} W^{(m)} h^{(m-1)}_i\Big) , \quad (4)$$

where $\sigma(\cdot)$ is the sigmoid activation function; $W^{(m)}$ is a learnable weight matrix for the linear combinations of the node features; $a^{(m)}_{ij}$ is the inter-node attention weight which quantifies the connection strength between node i and node j. Following the standard settings of GAT attention mechanism, when receiving an embedding h_j from a neighbor node j, node i performs self-attention to learn $a^{(m)}_{ij}$ as

$$a^{(m)}_{ij} = \mathrm{softmax}\big(f_{v_i}\big(W^{(m)}h^{(m-1)}_i, W^{(m)}h^{(m-1)}_j\big)\big) , \quad (5)$$

where f_{v_i} is a two-layer multilayer perceptron (MLP) with LeakyReLU activation. We note that $a^{(m)}_{ij}$ can be used as a reference to determine which nodes should be emphasized or ignored during the link scheduling. After multiple rounds of embedding information exchanges among neighboring nodes, the node representation of graph topology becomes more refined and coordinated, which is helpful for the global optimal resource allocation strategy. Then, for each G_t, we adopt an MLP as the readout function $\mathbf{R}_\theta(\cdot)$ to obtain the whole graph representation as

$$h^{(m)}_{\mathrm{G}_t} = \mathbf{R}_\theta\big(\{h^{(m)}_j | j \in \mathbb{V}^{(m)}_t\}\big) . \quad (6)$$

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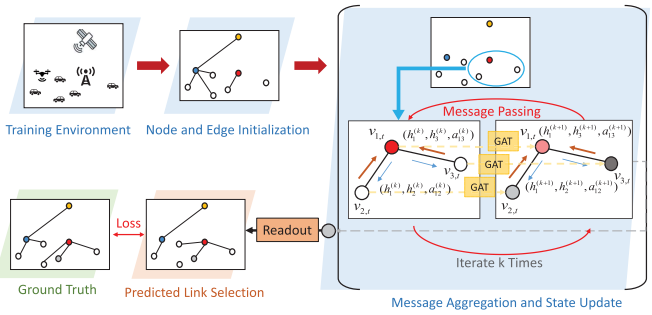


Fig. 2. An illustration of the graph processing of the proposed GAT model during the training phase.

By treating the link predication task as a supervised node classification problem, we adopt the negative log-likelihood loss function as

$$L = \sum_{v \in \mathbb{V}_t} y_v \log(\sigma(h^{\text{out}}_{\text{G}_t,v})) + (1 - y_v) \log(1 - \sigma(h^{\text{out}}_{\text{G}_t,v})) \ , \tag{7}$$

where y_v is the label[2] of node v indicating the ground-truth link selection; $h^{\text{out}}_{\text{G}_t,v} \in [0, 1]$ is the final feature representation of graph G_t for node v. We use $h^{\text{out}}_{\text{G}_t,v}$ to obtain the link selection vector $\mathbf{u_t}$. Specifically, the n_{th} user is connected to the k_{th} UAV (i.e., $\alpha_{k,n,t} = 1$ and $\beta_{n,t} = 0$) if its final feature representation $h^{\text{out}}_{\text{G}_t,v}$ is greater than 0.5, otherwise the user is connected to the MBS (i.e., $\alpha_{k,n,t} = 0$ and $\beta_{n,t} = 1$).

After obtaining the global link scheduling result, each UAV decides its individual movement based on its local observations. Specifically, we use the obtained node-embedding features of the whole graph as part of the state vector in the Q-learning agent. By doing so, each UAV can take the link selection into consideration when making flight decisions. We note that the satellite index s can be ignored hereafter since the global link selection has been determined. The learning process can be formulated by the Markov decision process (MDP) and expressed as a four-tuple $(\mathbf{S}, \mathbf{A}, \mathcal{R}, \gamma)$, where:

- $\mathbf{S}$ denotes the state space. The state of the k_{th} UAV, denoted by $\mathbf{s}_k$, is described by its location, embedding features, and energy consumption, i.e., $\mathbf{s}_k = (w^{\text{UAV}}_{k,t}, h^{\text{out}}_{\text{G}_t,k}, E^{\text{fly}}_{k,t})$. To constrain the state space in a finite space, we discretize the entire area into cubes of equal length and the accumulated energy consumption into N_E levels.
- $\mathbf{A}$ denotes the action set of all the available movement at a given state. The action of the k_{th} UAV, denoted by $\mathbf{a}_{k,t} = (\theta_{k,t}, l_{k,t})$, consists of its direction $\theta_{k,t}$ and moving distance $l_{k,t}$. Here, we discretize the angle space of UAV's flight direction into N_A intervals and the moving distance per time slot into N_D levels.
- As per optimization problem (3a), the instantaneous reward received by the k_{th} UAV, due to taking the action $\mathbf{a_k}$, is defined as the coverage at time t, i.e., $\mathcal{R}_t = \sum_{n=1}^{N} U_n [\mathbf{a}_{k,t}, \mathbf{u_t}]$. The long-term cumulative reward for a sequence of states over T can be expressed as $R = \sum_{t=1}^{T-1} \gamma_t \mathcal{R}_t$, where $\gamma \in (0, 1]$ is the discount factor.

[2]We generate such ground-truth datasets by simulating the complex system. From these historical graph sequences, it is expected that the long-term heterogeneous relations between users and UAVs/MBS can be captured.

Algorithm 1 Proposed Algorithm

Input: $\mathbf{w}^{\text{LEO}}_{\mathbf{t}}$,$\mathbf{w}^{\text{U}}_{\mathbf{t}}$, $(x_b, y_b, 0)$, η_{th}, and E_{max}.
Output: $\mathbf{w}^{\text{UAV}}_{\mathbf{t}}$,$\mathbf{u_t}$

```
Set M = 0.
repeat
    Initialize u_t.
    Using the pre-trained GNN model to find u_t.
    Each UAV initializes Q_k(s, a_k).
    Each UAV chooses an action a_k based on the ε-greedy action
    selection mechanism and obtains the immediate reward R_t.
    Each UAV updates Q-values based on (8).
    Set M ← M + 1.
until M = M_max or convergence is achieved.
```

Through the standard Q-learning method, the optimal Q-value function can be achieved according to Bellman's principle of optimality. To be specific, we adopt the so-called Bellman update as follows

$$Q_k(\mathbf{s}, \mathbf{a}_k) \leftarrow (1 - \mu) Q_k(\mathbf{s}, \mathbf{a}_k) + \mu [\mathcal{R}_t + \gamma \max_{\mathbf{a}'_k} Q_k(\mathbf{s}', \mathbf{a}'_k)] \ , \tag{8}$$

where μ is the learning rate. To strike a balance between exploration and exploitation, we adopt the ϵ-greedy action selection in which the UAV selects a random action with probability ϵ and selects the action corresponding to the highest Q-value with probability $1 - \epsilon$.

The details of the two-stage learning algorithm are given in Algorithm 1. We start our algorithm by randomly generating the initial next positions of the UAVs. Then, we find the optimal link selection and the node-embedding features using the pre-trained GNN model which minimizes the loss function in (7). After that, each UAV interacts with the environment and then updates the Q-value function according to (8). This process is repeated until the policy converges or a maximum number of iterations M_{max} is reached. According to [11], to ensure the convergence of RL algorithms, the following two conditions need to be satisfied: 1) the state and action spaces are finite and 2) the learning rate μ_t satisfies $\sum_{t=0}^{\infty} \mu_t = \infty$ and $\sum_{t=0}^{\infty} \mu_t^2 < \infty$. Since all the state and action variables (including the embedding features obtained by GNN) are discretized, the first condition is satisfied. We set the learning rate to decay over time and hence the second condition is also satisfied. Therefore, the convergence of the proposed algorithm is ensured.

IV. Simulations

In this section, we evaluate the performance of the proposed learning algorithm (denoted by GNN-QL). We adopt the standard Q-learning (denoted by QL) and deep Q-network (denoted by DQN) as two baseline schemes for UAV trajectory design[3]. We consider an area of size 10 km $\times$ 10 km, where $N = 50$ ground users are randomly distributed. The orbit height of the LEO satellite is 780 km. The SR fading channel

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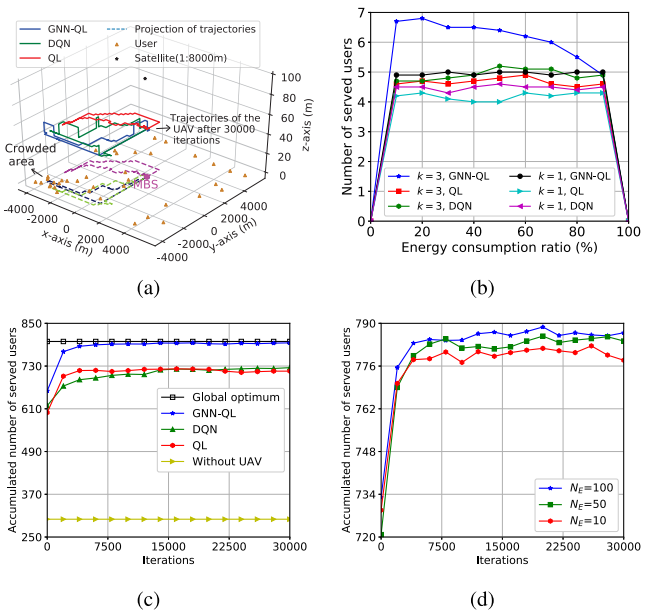


Fig. 3. (a) UAV trajectories. (b) Number of served users versus energy consumption ratio. (c) Accumulated number of served users. (d) Accumulated number of served users for GNN-QL with different discretization levels of energy consumption.

parameters are set as $(\Omega_{s,k}, b_{s,k}, m_{s,k}) = (0.0005, 0.063, 2)$. We fix the GNN model with two hidden layers. Unless otherwise specified, other settings are summarized as follows: $S = 1$, $P_{\text{LEO}} = 45$ dBm, $P_{\text{UAV}} = 20$ dBm, $P_{\text{BS}} = 40$ dBm, $N_0 = -110$ dBm, $\gamma_1 = \gamma_2 = 2$, $B = 1$ MHz, $\eta_{\text{th}} = 5$, $h_{\min} = 50$ m, $h_{\max} = 100$ m, $E_{\max} = 100$ Kilojoules, $N_E = 50$, $N_A = 4$, $N_D = 1$, $\gamma = 0.95$, and $\epsilon = 0.5$.

Fig. 3(a) plots the UAV trajectory and the user positions for $K = 1$ during the whole flight period. To simplify the graphic illustration, we plot only the initial positions of the users and the satellite. Recall that the start and end points of the UAV (in the horizontal plane) are both set at the position of the MBS. From the result, we can observe that the proposed GNN-QL allows the UAV to move to the crowded area more directly, as opposed to QL and DQN that fail to reach the center of the crowded area. Therefore, more ground users can be served by the UAVs under the energy budget constraint. Fig. 3(b) shows the number of served users versus energy consumption ratio with $K = 1$ and 3. We can observe that the proposed GNN-QL achieves the longest duration as well as the largest number of served users. This is expected since the global link selection result obtained by the proposed graph-based link scheduling can help UAVs to focus on the cell-edge users that cannot be served by the MBS. Fig. 3(c) shows the accumulated number of served users versus iteration number. It can be seen that the schemes with UAV trajectory control are significantly superior to the scheme without UAV relay, which further demonstrates the importance of UAV relaying. We also observe that the proposed GNN-QL achieves the fastest convergence rate and improves the coverage performance by 15% and 10% compared to QL and DQN, respectively. Moreover, GNN-QL provides a result that is very close to the global optimum solution obtained by the exhaustive search. Fig. 3(d) illustrates the accumulated number of served users for GNN-QL with different discretization levels of energy consumption. We can see that the accumulated number of served users significantly increases as the discretization level N_E increases from 10 to 50, while the performance improvement is not obvious when N_E increases from 50 to 100.

[3]For the two baseline schemes, we simplify the evaluation by restricting that each ground user is served by the nearest UAV/MBS. Accordingly, the node embedding feature $h^{\text{out}}_{\text{G}_t,k}$ is not included in the state vector of the standard Q-learning. In the case of multiple UAVs, we adopt the standard decentralized multi-agent reinforcement learning (MARL) framework [12] in which each agent has an independent Q table with no interaction between other UAVs.

V. Conclusion

In this letter, we have jointly optimized the link scheduling and 3D UAV trajectory in HSTN to maximize the number of served ground users. With the aid of GNN and RL techniques, a two-stage learning algorithm is proposed to tackle the challenging non-convex problem. To our best knowledge, this work is the first attempt to introduce graph embedding for link selection into learning-based UAV trajectory design. We have shown that the proposed GNN-QL scheme is advantageous over the standard QL and DQN schemes under the UAV energy budget constraint.

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